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Students Undergraduate Research Graduate Excellence (SURGE)

Simulation and Preliminary Study of Lithium-Thionyl Chloride Reserve Batteries for Short-Duration High-Reliability Applications

End-Term Report

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Abstract

This project presents the simulation and preliminary study of Lithium-based reserve batteries for short-duration high-reliability applications such as those required in artillery fuze. The idea is that the battery must remain inactive during storage and get triggered into action only when the fuze is deployed. This requires a careful study of the transient state evolution of the cell's properties as well as mechanical design that enables its intended functionality. I have used COMSOL Multiphysics 6.3 for simulation and Solidworks Design 2026 for CAD; both of which are available within IITK's list of softwares.

The work that I have done has five parts: the literature review of the lithium thionyl chloride electrochemistry [1] and reserve battery design [2]; two-phase multiphysics flow simulation of the liquid-gas interface; the electrochemical discharge attempted through COMSOL and eventually implemented through MATLAB; a preliminary CAD formulating a possible cell architecture on Solidworks Design and finally, some introductory exposure to the ESME laboratory's coin-cell manufacturing processes.

While I got good results on the electrolyte activation simulation, I ran into problems designing a physically reasonable electrochemical discharge chemistry given that COMSOL Multiphysics is natively suited to intercalation chemistries (as seen in lithium ion batteries) or aqueous stoichiometric chemistries (as seen in lead acid batteries). Neither of them can possibly be repurposed into the kind of reducible catholytic chemistry that lithium thionyl chloride cells feature. As such, we were motivated to explore alternative methods such as the MATLAB implementation which has low fidelity but produced a physically realistic simulation.

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Chapter 1

Introduction

1.1 Reserve Batteries and Primary Cells

A primary battery is one that is designed for a single discharge cycle. Once the active material is consumed, the cell can no longer be recharged and utilised to deliver power. This is in contrast to secondary batteries such as Li-ion batteries which operate upon the principle of repeated intercalation and de-intercalation and allow multiple charge-discharge cycling intervals. In our focus of study for this project, we focus on applications where the cell sits dormant for an incredibly long duration and must deliver power reliably at a moment's notice. This would lead to undesirable effects of capacity-fade and self-discharge that are ordinarily associated with the storage of a "live" secondary cell.

A reserve battery provides a novel mechanism to solve these concerns. In such batteries, we see that the active electrolytic material is kept physically isolated from the surrounding electrode stack. This ensures that any activation will be a consequence of a deliberate triggering action, upon which the previously isolated material would flood into the surrounding region into the stack to activate the cell. Naturally, this isolation eliminates the two dominant failure modes of long-term battery storage: (1.) self-discharge (since there is no closed electric circuit until activation) and (2.) material degradation due to prolonged contact between reactive species.

Consequently, reserve batteries can be seen as an optimal fit for two pivotal causes. One, in uses requiring long term storage that may even stretch up to a decade or beyond and two, in cases where we require urgent triggering for short-duration high-power applications. This is something that neatly lends to its ideal usage in military and rocketry applications including torpedoes, artillery fuzes etc.

1.2 Lithium as a Reserve Battery Chemistry

While discussing choices for anode materials, Lithium often appears as a very attractive choice. This is thanks to not only its position in the electrochemical series (it has the highest oxidation potential of any metal used in practical batteries) but also due to its nature as the lightest metal giving it a very high specific energy compared to say, some aqueous chemistries such as Pb-acid or Ni-Cd.

Historically, several lithium couples have been explored for potential application in primary cells, including lithium sulphur dioxide (Li-SO₂) and lithium vanadium pentoxide (Li-V₂O₅), but lithium thionyl chloride couple (Li-SOCl₂) appears as a clear winner given its highest energy densities amongst evaluated lithium primary chemistries [McCartney, Lund & Sturgeon, 1979].

To further motivate the point, the following are some major advantages associated with this couple:

- A very high theoretical specific energy (order of 1470 Wh/kg).
- Liquid thionyl chloride serves as both a cathode as well as an electrolyte, i.e., as a "catholyte", which greatly simplifies construction compared to solid state batteries as the solvent and cathode reducible species are composed of the same active material.
- The reaction between Lithium anode and thionyl chloride catholyte results in the formation of a passive LiCl film which prevents further self-discharge, hence regulating active material consumption.
- Wide operating temperature range, with cells demonstrated to function from around -40°C to $+70^{\circ}\text{C}$.

These properties make Li-SOCl₂ a natural candidate for reserve battery applications, and that is why, it is the chemistry adopted for this project.

1.3 Motivation for This Project

Despite the practical maturity and well-charted Li-SOCl₂ chemistry, the computational modelling of its application in a reserve battery remains relatively underexplored, particularly so in the transient periods during and after electrolyte activation. Most openly available simulation modules such as COMSOL Multiphysics 6.3's Battery Design module fails to deal with these stages and so there is a distinct need to look for workarounds and take inspiration from pre-existing primary cell formulations as well as utilise MATLAB simulations to create a more transparent modifiable model.

This project therefore set out with a clear multi factor objective:

1. To simulate electrolyte activation through a two-phase level set model.
2. To design a corresponding CAD that houses the entire cell and appropriately enables the activation of the electrolyte that spills over and into the electrode stack.
3. To investigate the extent to which existing COMSOL Multiphysics modules can be repurposed to resemble thionyl chloride chemistry.
4. To find out if it is indeed possible to formulate a good MATLAB equivalent model that can simulate the two reaction regimes.
5. To gain exposure to laboratory practices at the ESME Lab to better understand cell fabrication.

1.4 Scope and Structure of This Report

We will make a small note that in this project, no actual reserve battery was fabricated. The laboratory work was instead limited to the fabrication of a coin-cell which helped to understand not only cell manufacturing processes but also understand the underlying chemistry behind primary cells. Similarly, the scope of our simulation work is bounded: having succeeded in modelling electrolyte activation, we ran into a major road block when we were faced with modelling electrochemical discharge of primary Li-SOCl₂ cell, with the existing COMSOL ecosystem heavily optimised to work on secondary cell chemistries, a limitation that we have discussed in chapter 6. With that out of the way, we will further elaborate upon the mitigational work including but not limited to the formulation of a new, MATLAB model.

The rest of the report is organised as follows: chapter 2 deals with literature review with references to the citations [1], [2] and [3]. We then move onto the chapter 3 that deals with the underlying theory governing our battery chemistry, accounting for all the different equations and theories that have constituted our COMSOL and MATLAB simulations. Chapters 4 and 5 are a brief account of my journey through electrochemical discharge modelling, spanning across preliminary efforts on COMSOL and then going onto the successful implementation in MATLAB. Chapter 6 is an overview of the methodology that was undertaken while designing this project in terms of the software tools that were required. Chapter 7 addresses electrolyte activation simulation, successfully implementing a level set method to model the multiphase interface using mostly pre-existing COMSOL machinery. Then, in chapter 8, I have proposed the mechanical design for a possible artillery fuze cell architecture, inspired by Thales' patent [2] and Diehl Energy Products [3]. Finally, I present my brief account of exploratory lab work done at the ESME laboratory before rounding off this report with discussion regarding the results (chapter 10) and insights into possible future work (chapter 11) and conclusion (chapter 11).

Chapter 2

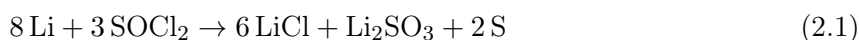
Literature Review

2.1 Lithium-Thionyl Chloride Electrochemistry

McCartney, Lund and Sturgeon’s 1979 review serves as a very useful foundational literature on Li-SOCl₂ cells, giving great arguments in favour of Li-SOCl₂ primary couples. Even though it’s been over 45 years since the paper was presented for the Naval Ocean Systems Center, it still remains as an invaluable relic of both the promise and historical safety concerns around lithium primary cells.

A Li-SOCl₂ cell is composed of a lithium metal anode, a porous carbon cathode current collector, a separator (Celgard, for e.g.), and an electrolyte that also acts as a cathode active species: a solute such as LiAlCl₄ dissolved in thionyl chloride (SOCl₂). The dual nature of the electrolyte, doubling as a reducible catholyte, is precisely what leads to a number of compatibility concerns centred around the electrochemical discharge process given the relative dearth of preexisting interfaces on primary cell architectures.

Two competing discharge reactions show up in the literature, depending on current density. At low discharge rates:



while at higher current densities the commonly accepted overall reaction is:



Both reactions produce solid products under normal conditions, which plays a huge role in making this chemistry far more viable than lithium-sulphur dioxide cells of the same era making it especially susceptible to venting and explosion accidents.

The chosen chemistry is also highly beneficial for enhancing the primary cell shelf life thanks to the passivation mechanism. On the first contact between the anode and the electrode, a thin LiCl layer forms on the anode and prevents further direct reaction during storage. This mechanism therefore lets the reserve batteries sit for long periods without degradation.

2.2 Performance Characteristics

We find that the reported performance figures for Li-SOCl₂ cells vary a lot with cell size and construction, but a few benchmark values are useful for grounding the modelling. The theoretical specific energy of the couple is around 1470 Wh/kg [1]. This number is a useful sanity check

against the MATLAB model’s predicted energy delivery in Chapter 5, even though the model here is not tied to any specific commercial cell geometry.

The voltage stability in these cells is quite notable. With the open circuit voltage quoted at 3.65 V, the typical loaded voltages are of the order of 3.2–3.4 V; the output voltage then remains constant-ish for nearly 95% of the cell’s life cycle before having a knee at the end characterising a steep fall. [1].

2.3 Reserve Battery Architectures

Generally, a reserve battery has three functional pieces: an electrode stack (comprised of anode, cathode and the separator arranged as per design requirements). The geometry varies a lot (planar stacks, wound jelly-roll configurations, annular stacks around a central reservoir), but the underlying principle of isolation followed by triggered activation is the same across all of them.

Amongst the many possible architectures, this project has chosen to build upon Thales’ artillery fuze configuration [2], which has been demonstrated in the Chapter 8, and for whom the entire framework has been primarily formulated.

2.4 Electrolyte Activation Physics

Once an electrolyte reservoir is opened, the electrolyte has to flow and fill the volume around the electrode stack before the cell can deliver meaningful current. The interface between the incoming electrolyte and whatever gas or vacuum previously occupied that space evolves under gravity, inertia, viscous drag, and surface tension. This is a two-phase, immiscible-fluid flow problem, and modelling it correctly means solving the Navier-Stokes equations for both fluid phases while tracking the moving interface between them. This is the approach taken in Chapter 7, using the Level Set method developed in Section 3.5.

This filling transient matters because it sets a lower bound on how quickly a reserve battery can start delivering rated current after activation: a cell whose electrolyte has not yet fully wetted the electrode stack shows a rising, rather than immediately available, current-carrying capability. Even a generic, geometry-independent treatment of this transient was a useful complement to the discharge-phase modelling in Chapters 4 and 5.

2.5 Prior Computational Modelling of Reserve and Primary Lithium Batteries

Compared with the large literature on modelling rechargeable lithium-ion cells, for which commercial finite-element tools like COMSOL’s Battery Design Module offer dedicated, well-validated physics interfaces, there is comparatively little published work on computationally modelling primary, reducible-catholyte reserve batteries such as Li-SOCl₂. Much of the published characterisation of this chemistry, including the McCartney, Lund, and Sturgeon review used throughout this chapter, is empirical and safety-focused rather than built around predictive multiphysics simulation [1]. That gap is what pushed this project toward a custom lumped-parameter model in MATLAB (Chapter 5), and it comes back up as a direction for future work in Chapter 11.

Chapter 3

Theory

3.1 The Nernst Equation

The equilibrium (open-circuit) potential of an electrochemical half-reaction is given by the Nernst equation. For a general reduction reaction



the electrode potential E relative to the standard potential E^0 is:

$$E = E^0 - \frac{RT}{nF} \ln \left(\frac{a_{\text{Red}}}{a_{\text{Ox}}} \right) \quad (3.2)$$

where R is the universal gas constant, T the absolute temperature, n the number of electrons transferred, F the Faraday constant, and a_{Red} , a_{Ox} the activities of the reduced and oxidised species. This describes the thermodynamic, zero-current cell voltage; drawing current adds overpotentials, the most significant of which for battery modelling is the activation overpotential from the Butler-Volmer equation below.

3.2 The Butler-Volmer Equation

At an electrode-electrolyte interface carrying net current, the local overpotential η and the resulting current density i are related by the Butler-Volmer equation:

$$i = i_0 \left[\exp \left(\frac{\alpha_a F \eta}{RT} \right) - \exp \left(-\frac{\alpha_c F \eta}{RT} \right) \right] \quad (3.3)$$

where i_0 is the exchange current density (how kinetically facile the reaction is at equilibrium), α_a and α_c are the anodic and cathodic charge transfer coefficients (usually $\alpha_a + \alpha_c = n$), and $\eta = E - E_{\text{eq}}$ is the overpotential, the deviation of the electrode potential from equilibrium.

The two exponential terms represent the forward (oxidation) and reverse (reduction) reaction rates; at $\eta = 0$ they cancel and no net current flows, which matches an electrode sitting at equilibrium. This equation is the standard kinetic boundary condition at electrode surfaces in COMSOL's Tertiary Current Distribution interface, and getting a physically reasonable i_0 for the Li-SOCl₂ couple turned out to be one of the practical difficulties in Chapter 4.

3.3 Species Transport: The Nernst-Planck Equation

Within the electrolyte, the flux of a charged species j has three contributions: diffusion (concentration gradients), migration (electric field), and convection (bulk fluid motion). This is the Nernst-Planck equation:

$$N_j = -D_j \nabla c_j - z_j u_j F c_j \nabla \phi + c_j \mathbf{v} \quad (3.4)$$

where N_j is the molar flux of species j , D_j its diffusion coefficient, c_j its concentration, z_j its charge number, u_j its ionic mobility, ϕ the electrolyte potential, and $\mathbf{v}$ the bulk fluid velocity. Most stationary-electrolyte battery models drop the convective term, but it matters during the activation transient of a reserve battery, when the electrolyte is physically moving to fill the cell. This is part of why the activation-phase and discharge-phase physics could not simply be bolted together in this project (see Chapter 4).

Combined with local electroneutrality, or more rigorously Poisson's equation for the electric field, this flux expression for each species gives species conservation in the electrolyte domain:

$$\frac{\partial c_j}{\partial t} + \nabla \cdot N_j = R_j \quad (3.5)$$

where R_j is a source or sink term for any homogeneous reactions producing or consuming species j in the bulk electrolyte, as opposed to at the electrode surface, which is handled through the boundary condition above.

3.4 Charge Conservation

Current continuity requires the divergence of the total current density (ionic current in the electrolyte, electronic current in the electrode) to vanish in the absence of accumulation:

$$\nabla \cdot i_l + \nabla \cdot i_s = 0 \quad (3.6)$$

where i_l is the ionic current density in the electrolyte and i_s the electronic current density in the solid electrode or current collector. The ionic current relates to the species fluxes by:

$$i_l = F \sum_j z_j N_j \quad (3.7)$$

These two equations, together with Ohm's law for the electronic phase ($i_s = -\sigma \nabla \phi_s$, with σ the electronic conductivity), close the system of equations solved in COMSOL's current distribution physics.

3.5 Two-Phase Flow and the Level Set Method

The electrolyte activation transient (Chapter 7) involves two immiscible fluids, the incoming electrolyte and the gas or vacuum it displaces, sharing a domain and separated by a moving interface. Within each phase, the flow follows the incompressible Navier-Stokes equations:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot [\mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)] + \mathbf{F} \quad (3.8)$$

$$\nabla \cdot \mathbf{u} = 0 \quad (3.9)$$

where $\mathbf{u}$ is fluid velocity, p pressure, ρ and μ the phase-dependent density and dynamic viscosity, and $\mathbf{F}$ any body force (gravity, and at the interface, surface tension).

The hard part of a two-phase problem is that ρ and μ are discontinuous across the interface, and the interface itself moves with the flow and has to be tracked as part of the solution. The Level Set method handles this with a smooth scalar function $\phi(\mathbf{x}, t)$ defined over the whole domain, where $\phi = 0$ marks the interface and $\phi > 0$, $\phi < 0$ mark the two phases. Rather than an integer indicator function, ϕ varies smoothly (typically a smoothed signed-distance-like function) over a thin transition band of width ε around the interface, which lets fluid properties be written as one smooth expression over the whole domain:

$$\rho(\phi) = \rho_1 + (\rho_2 - \rho_1)H(\phi), \quad \mu(\phi) = \mu_1 + (\mu_2 - \mu_1)H(\phi) \quad (3.10)$$

where $H(\phi)$ is a smoothed Heaviside function going from 0 to 1 across the interface band, and subscripts 1 and 2 denote the two fluid phases.

The interface is advected by the local flow velocity through a transport equation for ϕ :

$$\frac{\partial \phi}{\partial t} + \mathbf{u} \cdot \nabla \phi = \gamma \nabla \cdot \left(\varepsilon \nabla \phi - \phi(1 - \phi) \frac{\nabla \phi}{|\nabla \phi|} \right) \quad (3.11)$$

The right-hand side is a reinitialisation term rather than a physical effect: its job is purely numerical, keeping ϕ well-behaved across the interface band as the simulation runs, so the function does not become too steep or too flat over time, which would otherwise hurt the accuracy of the fluid property interpolation above. The parameters γ and ε control the reinitialisation rate and interface thickness, and were tuned to the mesh resolution used in Chapter 7.

Finally, a wetted-wall boundary condition was used in place of the standard no-slip condition at the container walls in Chapter 7. This lets the contact line, where the fluid-fluid interface meets a solid boundary, move along the wall, governed by a prescribed or computed contact angle. That is physically necessary for the electrolyte-filling problem, since a strict no-slip condition would pin the contact line in place and stop the electrolyte front from advancing along the container walls in a way that matches real wetting behaviour.

Chapter 4

Electrochemical Modelling in COMSOL

4.1 First Attempt: Adapting the Standard Lithium-Ion Battery Example

The starting point for electrochemical modelling was COMSOL’s built-in 1D isothermal lithium-ion battery example, in the Battery Design Module. This example ships with a complete, working three-layer cell geometry (negative electrode, separator, positive electrode, with layer thicknesses $L_{\text{neg}} = 100\ \mu\text{m}$, $L_{\text{sep}} = 52\ \mu\text{m}$, $L_{\text{pos}} = 174\ \mu\text{m}$ in the version used here), pre-configured particle-scale solid diffusion for each electrode, and a full Tertiary Current Distribution formulation with Butler-Volmer kinetics already implemented at each porous electrode.

The appeal of starting here was simple: rather than build a current-distribution model from scratch, an already-validated, already-converging model could in principle just be re-parametrised with Li-SOCl₂-relevant material properties and boundary conditions. In practice this turned out to be a deeper mismatch than a simple re-parametrisation job.

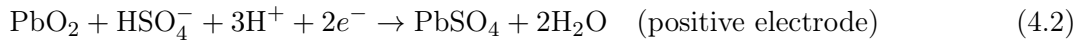
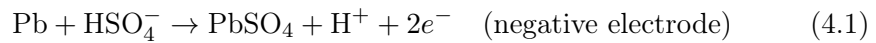
The lithium-ion example assumes both electrodes are intercalation hosts: the active material (graphite, NMC, etc.) hosts lithium ions within a solid particle, and the reaction at the particle surface is $\text{Li}^+ + e^- \rightleftharpoons \text{Li}(\text{host})$, a reversible insertion reaction tracked with a solid-phase diffusion equation inside each particle ($D_{s,\text{neg}} = 3.9 \times 10^{-14}\ \text{m}^2/\text{s}$, $D_{s,\text{pos}} = 1 \times 10^{-13}\ \text{m}^2/\text{s}$, particle radii $r_{p,\text{neg}} = 12.5\ \mu\text{m}$, $r_{p,\text{pos}} = 8\ \mu\text{m}$). Nothing about the electrolyte is consumed by this reaction: its role is purely to transport Li^+ ions between electrodes, and its salt concentration c_l only evolves through diffusion and migration, not bulk consumption.

Li-SOCl₂ chemistry is close to the opposite. As Chapter 2 laid out, thionyl chloride is the reducible species itself: it is irreversibly consumed at the cathode and converted into solid LiCl and other products, and there is no host material with a bounded, cyclable lithium capacity on the cathode side. Forcing this chemistry into the lithium-ion example’s framework, even after swapping in Li-SOCl₂-relevant electrode potentials and exchange current densities, meant the simulation was still solving for intercalation-style solid-phase diffusion into a particle that has no physical meaning for this chemistry. Running the modified model, the electrolyte and solid-phase concentration profiles stayed essentially flat over the whole discharge. The model converged numerically without any error, but it was not actually representing the depletion of the reducible catholyte that governs real Li-SOCl₂ discharge behaviour. A converged solution is not the same thing as a physically meaningful one. That was the first clear lesson of the project’s modelling phase.

4.2 Second Attempt: Repurposing the Lead-Acid Battery Interface

After that, the approach changed: instead of continuing to adapt the lithium-ion interface, the Lead-Acid Battery physics interface became the new starting point (kept in the project files as the LiSOCl2bat physics node, Lead_acid_battery type). This choice made more physical sense. In a lead-acid cell, the electrolyte (sulphuric acid) is a direct, stoichiometric participant in the electrode reactions: it is consumed and produced at the electrodes, and its concentration genuinely tracks depth of discharge. That structural feature, electrolyte as reactant, is much closer to how Li-SOCl₂ behaves than the lithium-ion module's electrolyte-as-transport-medium framework, which is why the switch made sense.

The lead-acid interface's electrode reactions are built around the specific stoichiometry of the lead-acid discharge reactions:



and exposes each species' stoichiometric coefficient (ν_{H^+} , $\nu_{\text{HSO}_4^-}$, $\nu_{\text{H}_2\text{O}}$, ν_{Pb} , ν_{PbO_2} , ν_{PbSO_4}) as an editable parameter at each porous electrode reaction node. The repurposing strategy here was to keep only a single generic charge-transfer species: setting $\nu_{\text{H}^+} = +1$ at the negative electrode and $\nu_{\text{H}^+} = -1$ at the positive electrode to stand in for a generic single-electron transfer analogous to Li⁺ release and consumption, and zeroing out every other lead-acid-specific coefficient to remove the lead-acid-specific solid products from the picture. Material properties were replaced throughout: the electrolyte was redefined with SOCl₂-like properties (density 1600 kg/m³, ionic conductivity 1.8 S/m, salt diffusivity 2.0×10^{-10} m²/s), and the porous carbon cathode was given an equilibrium potential of 3.65 V (the correct open-circuit value for Li-SOCl₂ from Section 2.1) in place of the lead-acid PbO₂ electrode's reference potential.

This attempt was a real improvement. With the reaction able to draw down the electrolyte salt concentration variable c_l , instead of c_l being decoupled from the reaction entirely as in the lithium-ion attempt, the resulting discharge curve had a much more physically plausible shape: a long, flat voltage plateau near the target 3.65 V equilibrium potential, matching the genuinely flat discharge characteristic that real Li-SOCl₂ cells show through most of their life (Section 2.2). But the model did not degrade gracefully at the end of discharge. Instead of the voltage falling off smoothly as depth of discharge approaches 100%, the solver plateaued and then aborted before completing the run.

4.3 Root Cause

Together, these two observations point to a specific, identifiable limitation rather than a vague "it didn't work." By zeroing out every lead-acid-specific stoichiometric coefficient except a single generic ν_{H^+} , the modified reaction no longer has any term tying electrolyte salt concentration c_l to a physically bounded, gracefully-vanishing reactant supply, the way the real reaction in Equation 2.1 does. The modified interface keeps the numerical machinery for coupling concentration to reaction rate, but not a stoichiometrically consistent picture of what is actually being consumed. As the true electrolyte concentration, or the porosity term ϵ (which the Lead-Acid Battery interface also evolves as a function of reaction extent, since the lead-acid reaction changes solid volume at the electrode), approaches a numerically small or non-physical value near the end of discharge, terms in the Butler-Volmer kinetics or the electrolyte conductivity expression (which depend on concentration or porosity, sometimes through a Bruggeman-type

exponent) become ill-conditioned. That is consistent with a solver that holds a plateau and then aborts, rather than one that predicts a physically graceful capacity fade.

This is a more precise version of the general difficulty flagged in Sections 1.3 and 2.5: COMSOL's Battery Design Module physics interfaces are built around specific, named electrochemical systems (Li-ion intercalation, lead-acid), and their internal reaction bookkeeping is tied to those systems' specific stoichiometries. Swapping material properties is comparatively easy. Swapping the underlying reaction stoichiometry while keeping the model numerically well-posed at full discharge is considerably harder, and this project did not fully manage it.

4.4 What Would Be Needed to Do This Properly

Based on that diagnosis, a more principled approach (left for future work in Chapter 11) would be either (a) using COMSOL's more general Electrode Reaction and Electrolyte building-block physics, rather than a pre-packaged battery-chemistry interface, to define the true Li-SOCl₂ stoichiometry of Equations 2.1/2.2 from first principles, with a properly bounded, vanishing reactant supply term for SOCl₂ itself, or (b) accepting the lumped-parameter approach of Chapter 5 as the main modelling tool for this chemistry and keeping finite-element modelling for the electrolyte-activation fluid dynamics alone (Chapter 7), where COMSOL's general-purpose two-phase flow physics is not tied to any particular battery chemistry and worked as intended.

4.5 Transition to the MATLAB Model

Given how hard it was to get a solver-stable, physically meaningful discharge curve out of either COMSOL attempt, the decision was to build the lumped-parameter MATLAB model described in Chapter 5. This was not a workaround adopted lightly (losing spatial resolution is a real limitation, covered in Section 5.5), but it allowed the project to get a complete, well-behaved discharge curve grounded in the same underlying electrochemical theory (Chapter 3), without depending on a finite-element physics interface whose internal assumptions did not match the chemistry being studied.

Chapter 5

MATLAB Model

5.1 Motivation

As Chapter 4 covers, the COMSOL Battery Design Module was hard to adapt correctly to Li-SOCl₂ chemistry. Its built-in physics interfaces assume intercalation electrodes with a conserved, cyclable solid-phase lithium concentration, which does not map cleanly onto a primary cell where the cathode reaction irreversibly consumes the electrolyte itself. Rather than keep fighting the built-in module, a lumped-parameter discharge model was built independently in MATLAB. This trades away spatial resolution (the model has no explicit geometry or spatial concentration gradients) for a framework that is fully transparent, where every term ties directly back to the physics in Sections 3.1–3.4.

5.2 Model Architecture

The model treats the cell as a single lumped electrical and thermal node, updated at each time step by Coulomb counting. At every step k , the following are computed in order:

1. The applicable discharge reaction regime (Equation 2.1 or 2.2), chosen by comparing the instantaneous current density $I/A_{\text{electrode}}$ against a threshold value.
2. A temperature-corrected exchange current density, from an Arrhenius correction to the room-temperature value i_0 :

$$i_0(T) = i_{0,\text{ref}} \exp \left[-\frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{amb}}} \right) \right] \quad (5.1)$$

3. An open-circuit voltage that decreases linearly with depth of discharge and is corrected for temperature with a fixed coefficient dV/dT :

$$V_{\text{OCV}} = V_{\text{OCV,full}} - 0.15(1 - \text{SOC}) + \frac{dV}{dT}(T - T_{\text{amb}}) \quad (5.2)$$

4. A LiCl passivation resistance that grows as the cell discharges, following an empirical saturating exponential form:

$$R_{\text{pass}} = R_{\text{pass,max}} [1 - \exp(-k_{\text{pass}}(1 - \text{SOC}))] \quad (5.3)$$

This form was chosen to qualitatively match the fact that the passivating LiCl layer is thin, and so low-resistance, near full charge, since it has had little contact time or discharge current through it, and thickens as discharge proceeds.

5. The activation overpotential η , found by solving the Butler-Volmer equation for η given the applied current. This is a transcendental equation in η , so it is solved numerically with Newton-Raphson iteration at every time step.
6. The terminal voltage, combining the OCV, activation overpotential, and ohmic drop across the bulk electrolyte and passivation resistances:

$$V_{\text{term}} = V_{\text{OCV}} - \eta - I(R_{\text{bulk}} + R_{\text{pass}}) \quad (5.4)$$

7. Coulomb counting to update the state of charge, plus a lumped thermal balance combining ohmic and activation heat generation against convective loss to ambient, used to update the cell temperature for the next step's Arrhenius correction.

The simulation stops when V_{term} reaches the cutoff voltage (2.0 V), or when the maximum simulation time is reached, whichever comes first.

5.3 Assumptions and Simplifications

A few simplifications were necessary to make the model tractable at the lumped-parameter level:

- **No spatial resolution.** The model has no concept of position within the cell; concentration and potential are treated as spatially uniform. This is a stronger assumption than even the failed COMSOL attempt, which at least discretised the cell into layers.
- **Empirical rather than mechanistic passivation and OCV curves.** The equations above are fitted functional forms chosen to reproduce qualitatively correct discharge behaviour (falling voltage, rising internal resistance), not derived from first-principles solid-state diffusion through the LiCl layer. A more rigorous treatment would model LiCl layer growth kinetics explicitly.
- **Single lumped thermal node.** Internal temperature gradients within the cell are ignored; the whole cell mass is assumed to sit at one uniform temperature at each instant.
- **Fixed reaction regime per run.** The discharge reaction (Equation 2.1 vs. 2.2) is chosen once, at the start of the run, based on applied current density, and does not switch mid-run even if conditions change. That is a reasonable simplification for constant-current discharge, but it would need revisiting for a variable-load scenario.

5.4 Results

The model was run at a constant discharge current of $I = 0.20$ A against a nominal 2.0 Ah cell (a current density of 0.087 mA/cm² given the modelled electrode area, which puts the cell in the low-current-density reaction regime of Equation 2.1). Over a simulated window of 10,000 s (166.7 minutes), the cell did not reach the 2.0 V cutoff voltage; discharge proceeded from a state of charge of 100% down to roughly 55% over the simulated window.

Two things from this run are worth recording plainly rather than smoothing over:

The simulation window was too short for a full discharge at this current. A 2.0 Ah cell discharged at a constant 0.2 A would, in the idealised case, take about 36,000 s (10 hours) to reach full depth of discharge, considerably longer than the 10,000 s window used here. The

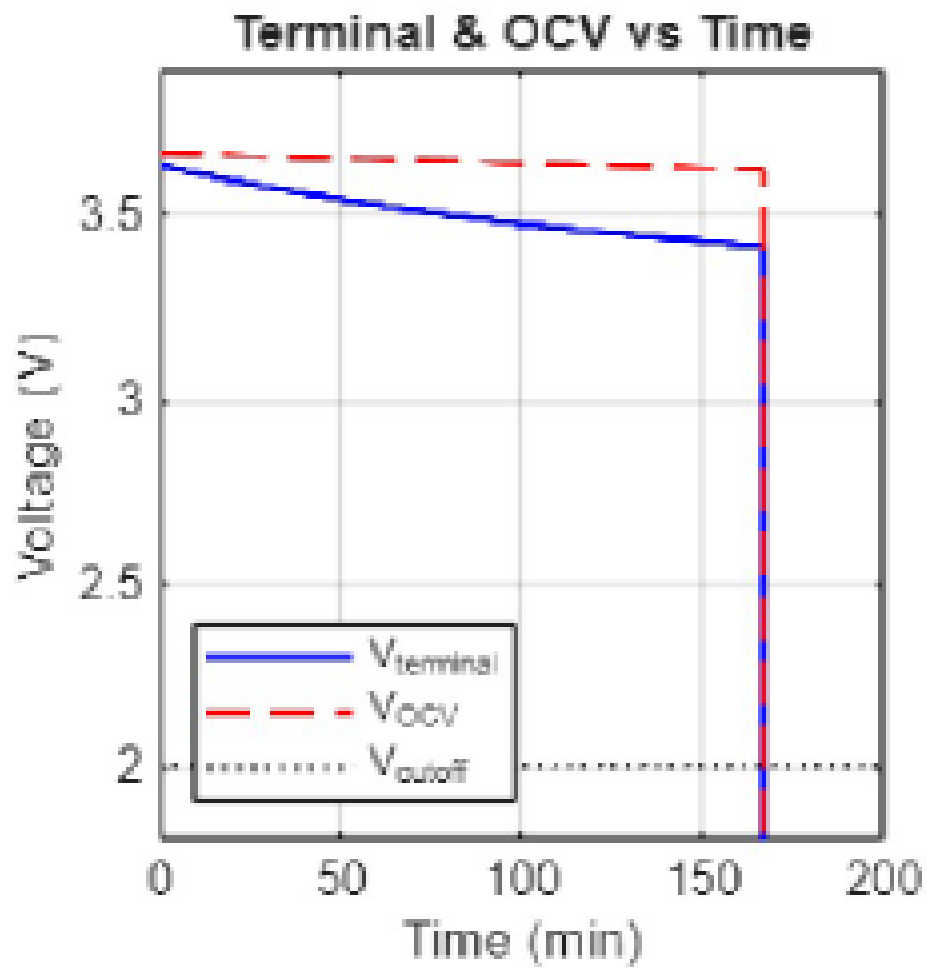


Figure 5.1: Terminal voltage and OCV vs. time

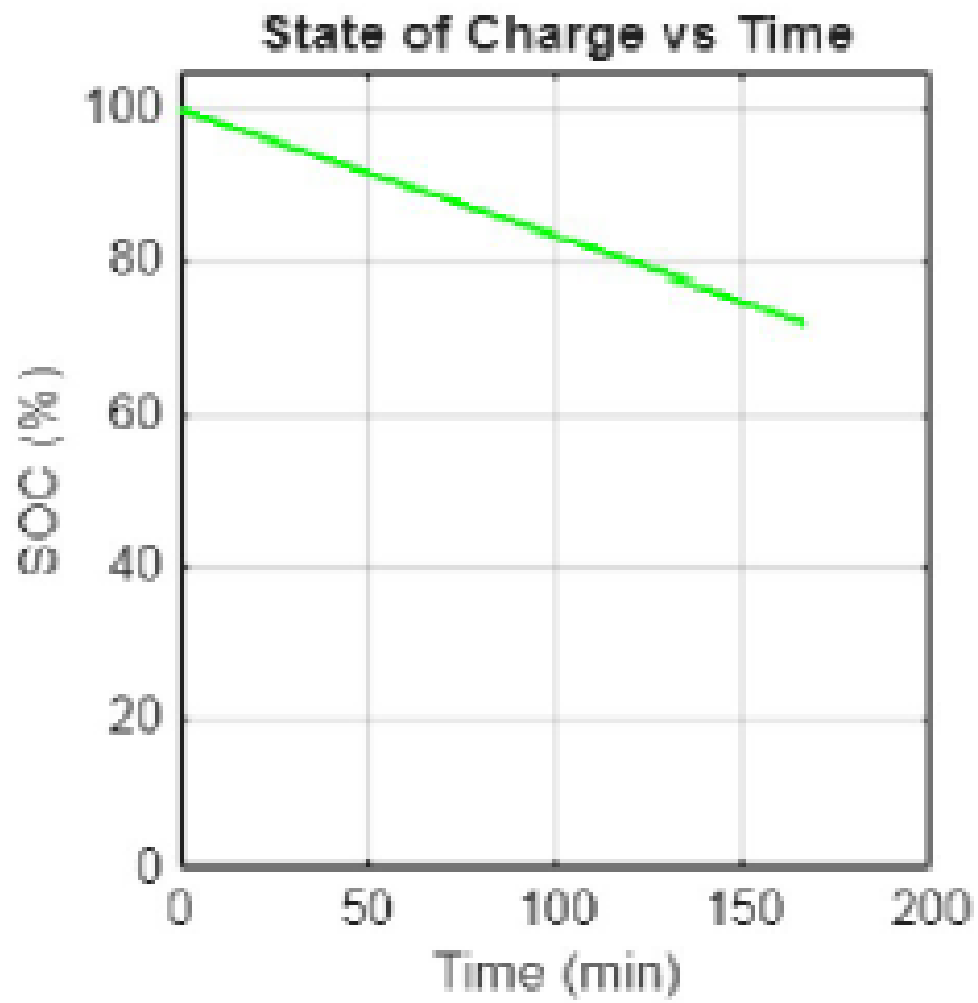


Figure 5.2: State of charge vs. time

results above should therefore be read as a partial-discharge snapshot rather than a complete discharge curve; running the same model out to the full 36,000 s is a direct extension and would need no changes beyond the simulation time limit.

An off-by-one indexing artefact in the script’s summary output. The main simulation loop fills $V_term(k)$ and $R_pass(k)$ for $k = 1$ to $nsteps - 1$, but the very last array index is never assigned inside the loop, since these values are computed from the state at step k before advancing to $k + 1$. As a result, the script’s end-of-run summary print statement, which reads $V_term(end)$, reports 0.0000 V instead of the true final computed voltage of 3.4165 V. This was only caught by inspecting the full time series rather than trusting the printed summary, which is a useful reminder to check the actual output rather than a script’s own summary line. The fix is simple: either log $V_term(k)$ one iteration earlier, or extend the loop to n_steps and compute a final half-step.

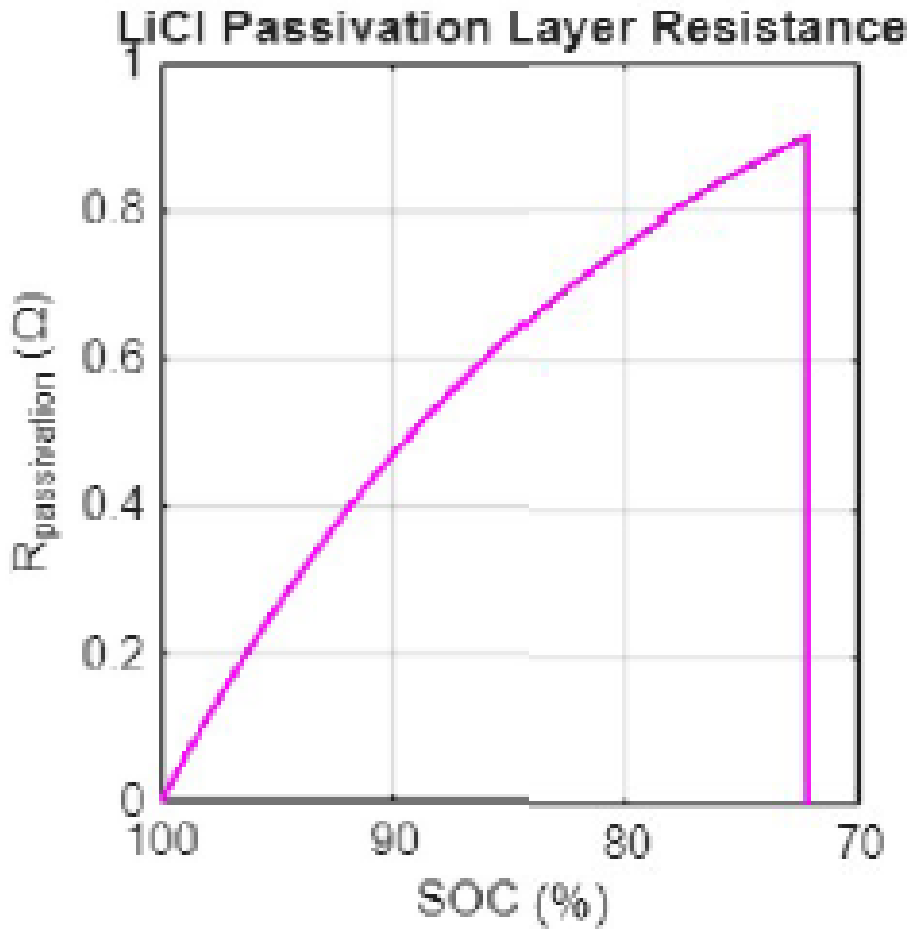


Figure 5.3: Passivation resistance vs. state of charge

5.5 Advantages and Limitations Relative to the COMSOL Attempt

The main advantage of the MATLAB model is that every term is fully under the author’s control and traceable back to the theory in Chapter 3, unlike the COMSOL Battery Design Module’s built-in physics, whose internal assumptions about electrode-electrolyte coupling turned out to be incompatible with a reducible-catholyte chemistry (Section 4.3). The trade-off is a complete

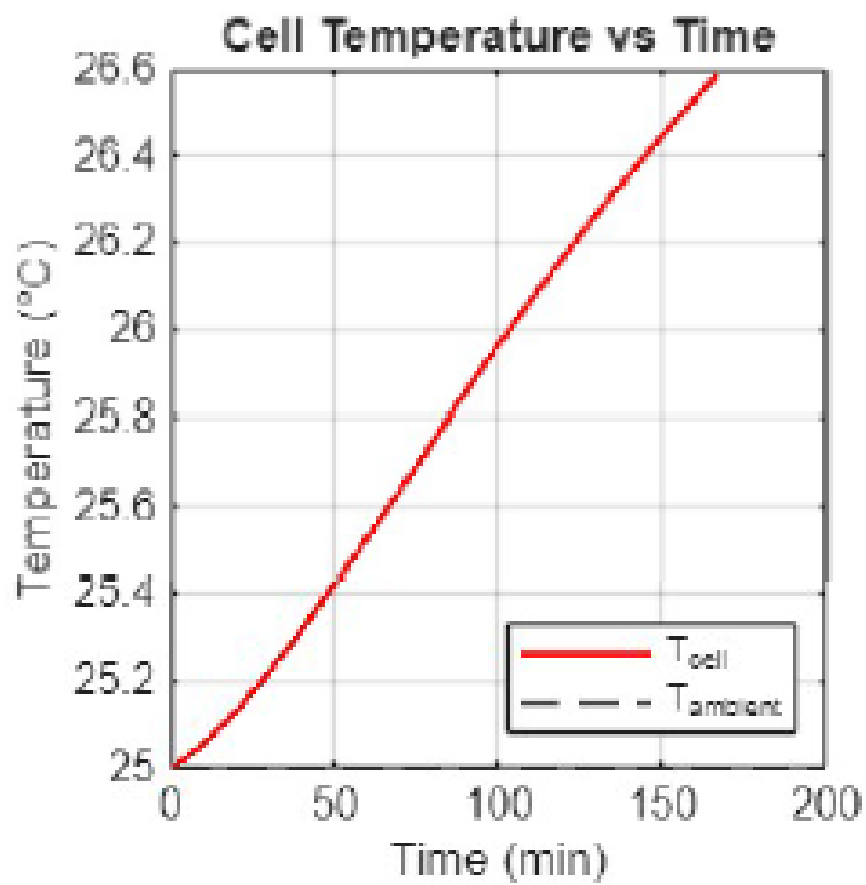


Figure 5.4: Cell temperature vs. time

loss of spatial resolution: the MATLAB model cannot predict concentration gradients across electrode thickness, or capture effects from non-uniform current distribution, both of which a correctly-configured finite-element model would in principle capture. The MATLAB model is better understood as a complementary, systems-level tool rather than a replacement for spatially-resolved multiphysics modelling. Ideally, a future version of this project would use the lumped model's parameters as a starting point for a correctly-formulated finite-element treatment (see Chapter 11).

Chapter 6

Methodology

6.1 Overall Workflow

The project moved through four broadly sequential, though overlapping, phases. The first was a literature review (Chapter 2), covering Li-SOCl₂ electrochemistry, reserve battery concepts, and the fluid mechanics of electrolyte activation, which fed both the theory in Chapter 3 and the physical parameters used throughout the modelling work.

The second phase was computational: first an attempt to model electrolyte activation using COMSOL's two-phase flow and Level Set physics (Chapter 7), then an attempt to model the electrochemical discharge that follows, first by adapting the built-in lithium-ion battery example and then by repurposing the Lead-Acid Battery interface (Chapter 4). When neither produced a numerically stable full discharge curve, a lumped-parameter model was built independently in MATLAB (Chapter 5).

Alongside the electrochemical modelling, a conceptual mechanical design phase used SolidWorks to explore a generic reserve-cell housing and electrode-stack arrangement at a preliminary CAD level (Chapter 8).

The final phase was hands-on laboratory exposure, observing the fabrication of standard laboratory coin cells (Chapter 9), meant to ground the computational work in a physical understanding of how lithium battery cells are actually assembled, even at a smaller and simpler scale than a reserve battery.

6.2 Software and Tools

- **COMSOL Multiphysics 6.3**, with the Battery Design Module, used for the two-phase electrolyte activation simulation (Chapter 7) and the electrochemical discharge modelling attempts (Chapter 4).
- **MATLAB**, used for the lumped-parameter discharge model (Chapter 5).
- **SolidWorks**, used for the conceptual mechanical CAD work (Chapter 8).

Both COMSOL and SolidWorks were accessed through IIT Kanpur's institutional software licenses.

Chapter 7

Electrolyte Activation Simulation

7.1 Motivation

As Section 2.4 covers, the process of an electrolyte filling the void around an electrode stack after activation is a genuine two-phase fluid dynamics problem, and its timescale places a practical lower bound on how fast a reserve cell can start delivering current. What follows is a generic, geometry-independent simulation of that filling process, built on the two-phase flow and Level Set theory from Section 3.5.

7.2 Geometry and Setup

A simplified 2D axisymmetric geometry represented a generic cylindrical reservoir: 2 cm wide by 1 cm tall, with the two fluid phases initially separated by a horizontal interface at height $h = 0.5$ cm. This geometry was chosen for simplicity: enough to capture the essential physics of a gravity- and inertia-driven interface evolution without the added complexity of a full 3D reserve-cell geometry, which was not necessary for validating the two-phase flow approach itself.

The Laminar Two-Phase Flow, Level Set multiphysics interface in COMSOL combined the Navier-Stokes formulation from Section 3.5 with the Level Set interface-tracking approach described there. The wetted wall boundary condition was used at the container walls in place of the standard no-slip condition, so the contact line could move realistically along the boundary as the interface evolved, rather than staying pinned.

7.3 Mesh and Solver

A time-dependent study resolved the transient evolution of the interface, velocity field, and pressure distribution.

7.4 Results

The simulation captured the qualitative behaviour expected of a gravity-driven two-phase interface: the velocity distribution, pressure field, and volume fraction (interface position) all evolved smoothly over the time-dependent study, without the numerical instability or non-convergence problems that affected the electrochemical modelling work in Chapter 4.

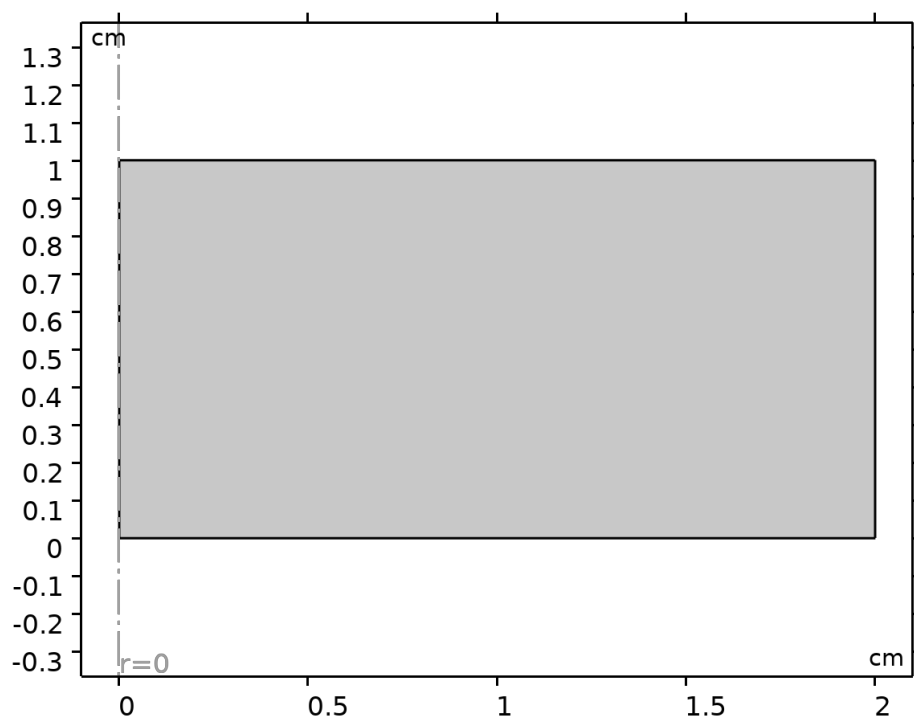


Figure 7.1: Simulation geometry

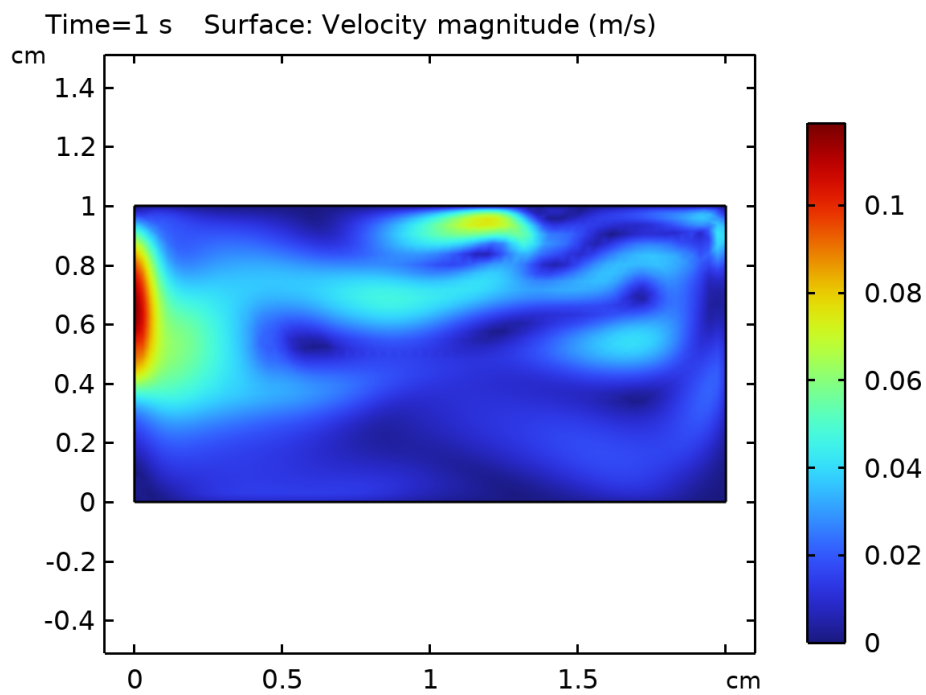


Figure 7.2: Velocity field evolution

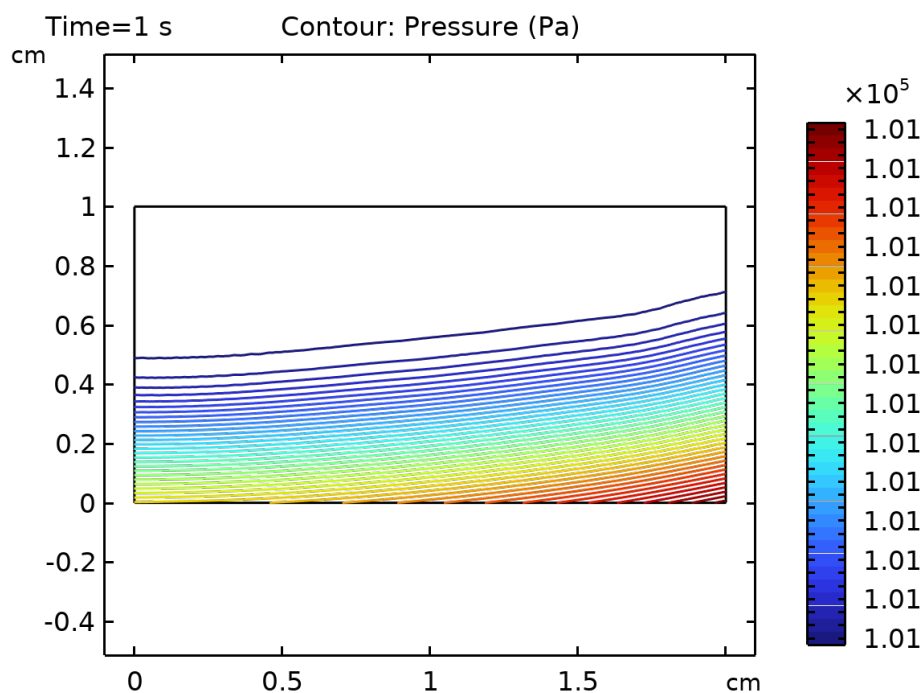


Figure 7.3: Isobaric pressure contours

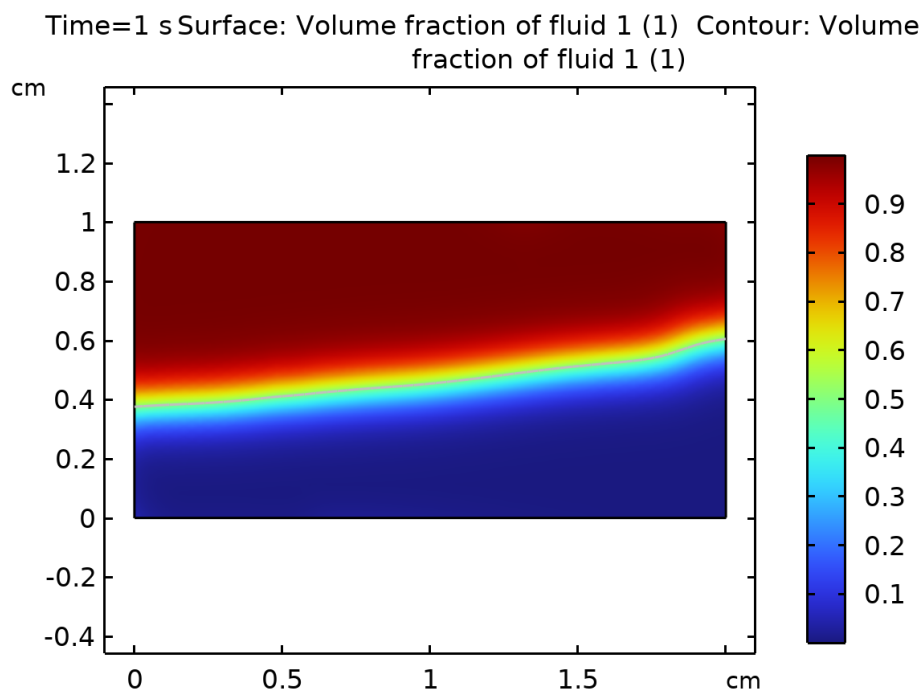


Figure 7.4: Volume fraction / phase evolution

This simulation working, compared with the electrochemical modelling in Chapter 4, is itself informative: COMSOL's general-purpose two-phase flow physics is not tied to any particular chemical system, so it did not need any chemistry-specific reaction stoichiometry to be specified correctly. This is unlike the Battery Design Module's physics interfaces, whose internal assumptions were the central obstacle in Chapter 4. Chapter 10 comes back to this distinction.

Chapter 8

Mechanical Design

8.1 Inspiration

The general design philosophy for the conceptual housing drew on Straver’s reserve battery patent [2] and on the reserve battery architecture described on Diehl Energy Products’ website [3].

8.2 Design Description

A conceptual reserve-cell housing, electrode stack, central cell ampoule (for storing the catholyte), and a plunger mechanism were designed to suit the use case required in artillery fuzes. The principal housing dimensions adopted for the design are listed in Table 8.1.

Table 8.1: Principal dimensions of the reserve battery housing.

Parameter	Symbol	Value
Housing outer diameter	D_{housing}	45 mm
Housing wall thickness	t_{wall}	1.5 mm
Electrode stack inner diameter	D_{stack}	28 mm
Ampoule outer diameter	D_{ampoule}	25.5 mm
Single-cell height	h_{cell}	1.66 mm

8.3 Individual Components

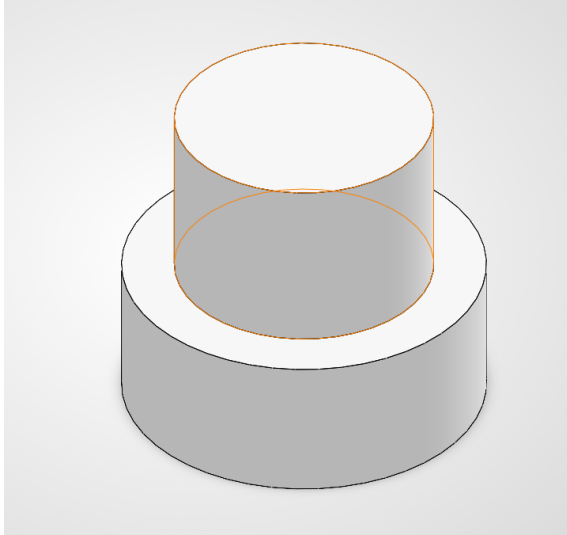


Figure 8.1: Housing

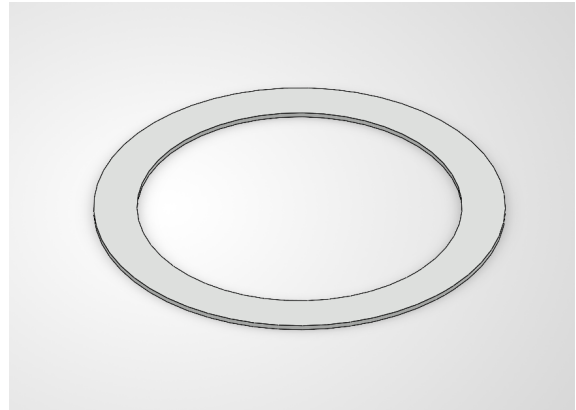


Figure 8.2: Electrode ring

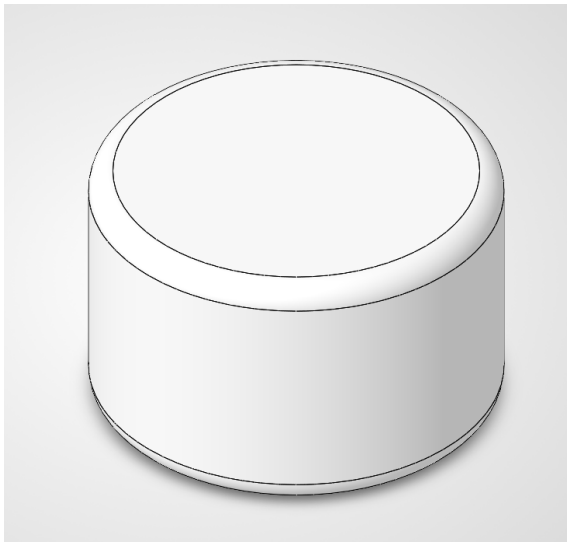


Figure 8.3: Electrolyte ampoule

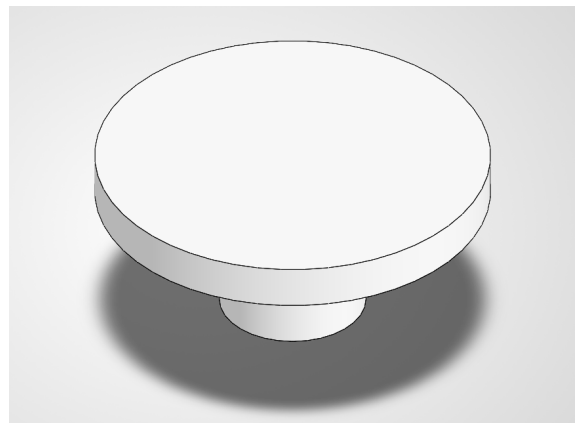


Figure 8.4: Plunger

8.4 Complete Assembly

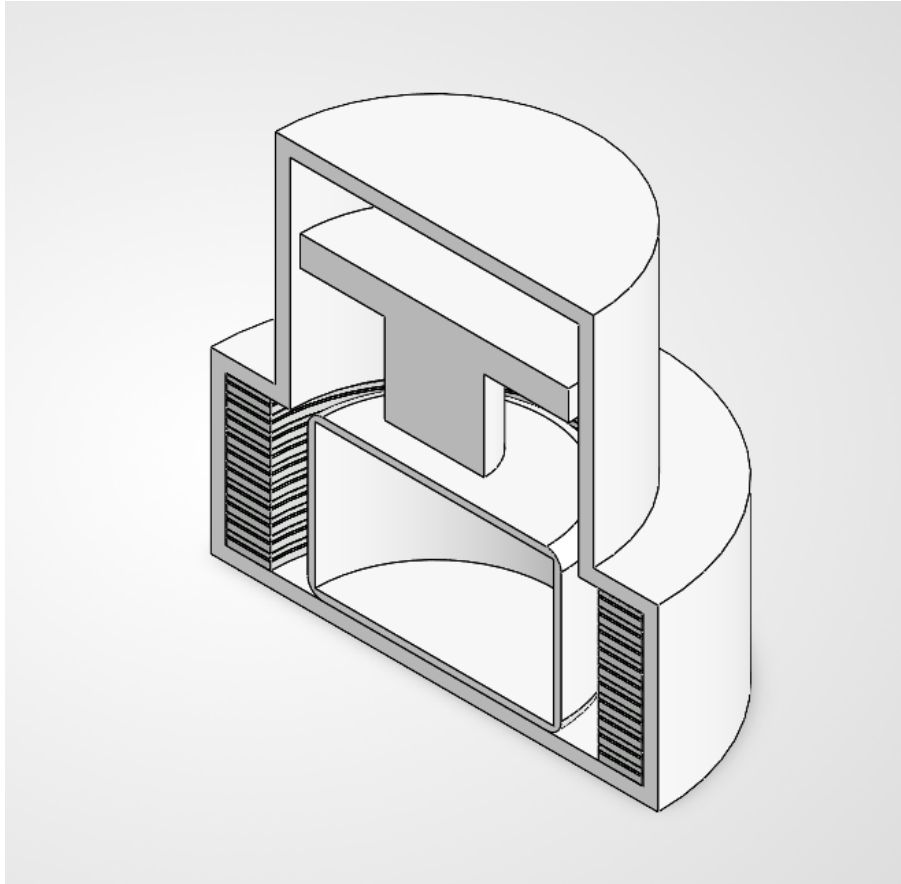


Figure 8.5: Exploded (or assembled) CAD model of the conceptual reserve battery.

8.5 Discussion

The housing serves as the structural enclosure for the reserve battery whereas the electrodes are arranged in a concentric stack with a consistent cathode-separator-anode stack separated by the means of a bipolar plate. The plunger mechanism is meant to be actuated by the means of some mechanical or inertial stimuli that causes it to crash into the glass ampoule thereby breaking it and spilling the catholyte all over the electrode stack. This activates the cell and essentially brings it to life.

Chapter 9

Laboratory Work

9.1 Motivation

To ground the computational modelling work in Chapters 4–7 in a physical understanding of how lithium battery cells are actually built, some lab time went into observing the fabrication of standard laboratory coin cells. As noted in Chapter 1, no reserve battery of any kind was fabricated during this project. The coin cells here are a standard laboratory format used broadly in battery research for its simplicity and reproducibility, and served purely as an educational, observational exercise.

9.2 Fabrication Steps Observed

The fabrication steps may best be represented by the following flowchart:

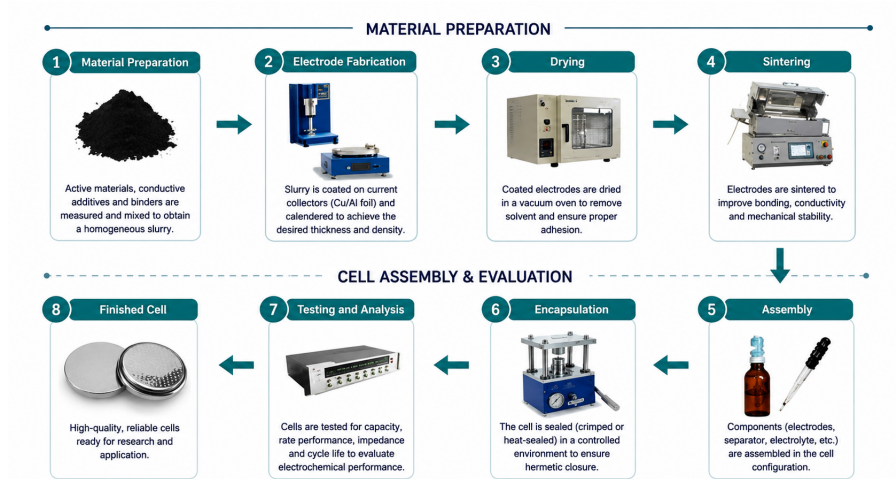


Figure 9.1: A flowchart briefly summarising different stages of coin-cell fabrication

Indeed, the fabrication process observed followed the general steps common to coin-cell assembly across much of the battery research literature:

- **Electrode preparation:** Active carbon cathode and lithium anode; for electrode preparation, we used an NMP + PVDF slurry coating on copper strip
- **Separator placement:** Celgard Separator

- **Cell stacking:** order of components within the coin cell casing
- **Crimping:** sealing the cell casing under controlled pressure

9.3 Experimental Data

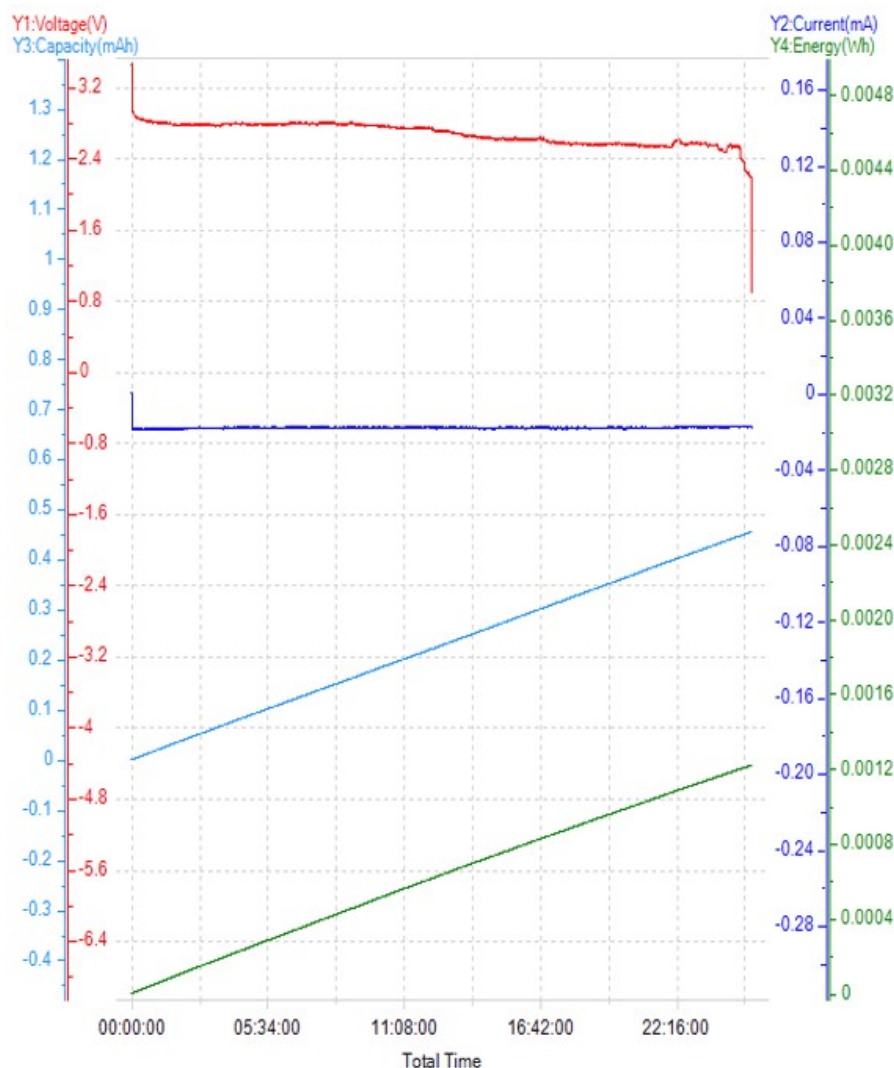


Figure 9.2: Initial testing results of the lab-fabricated cell

9.4 Relevance to Future Reserve Battery Development

Although the cells fabricated here were standard laboratory coin cells rather than reserve-type batteries, the fabrication techniques observed (controlled electrode preparation, separator handling, sealed cell assembly) carry over directly to any future reserve battery development effort. A reserve battery's electrode stack, once its electrolyte reservoir is activated, has to perform according to the same fundamental electrochemical principles as a standard primary cell. The real challenge in reserve battery design sits mainly in the isolation and activation mechanism (Chapter 8), not in the electrode-stack electrochemistry itself, which the coin-cell work here provided direct exposure to.

Chapter 10

Discussion

10.1 Comparing the Modelling Approaches

The four strands of technical work in this project, COMSOL two-phase flow (Chapter 7), COMSOL electrochemical modelling (Chapter 4), the MATLAB lumped model (Chapter 5), and the CAD and laboratory work (Chapters 8–9), land at quite different points on the trade-off between physical fidelity and practical tractability.

The two-phase flow simulation (Chapter 7) was the most straightforwardly successful piece of computational work in this project, precisely because it did not require adapting a chemistry-specific physics interface: the underlying Navier-Stokes and Level Set formulation is general-purpose, and needed no reaction stoichiometry to be specified.

The electrochemical modelling attempts (Chapter 4) were, by contrast, the most instructive failure of the project. Both the lithium-ion and lead-acid adaptations inherited internal assumptions from their source chemistries that did not map cleanly onto Li-SOCl_2 's reducible-catholyte behaviour: the first left concentrations essentially inert, the second produced a physically plausible plateau but an unstable end-of-discharge transition. The distinction between a model that "does not converge" and one that "converges but does not mean what you think it means" is one of the more useful practical lessons from the project.

The MATLAB model (Chapter 5) traded away spatial resolution entirely for full transparency and stability, and it produced a complete, well-behaved discharge curve grounded in the same theoretical framework (Chapter 3). Its main weakness is that several of its functional forms (the OCV-vs-SOC relationship, the passivation resistance growth law) are empirically fitted rather than derived from first-principles transport theory within the cell.

10.2 Strengths and Weaknesses, Summarised

Strengths: a working, validated two-phase activation simulation; a complete, internally consistent lumped discharge model with a clearly documented theoretical basis; a specific, evidence-based diagnosis of why two distinct COMSOL adaptation strategies each fell short in different ways; direct laboratory exposure to lithium cell fabrication.

Weaknesses: no finite-element model was obtained that correctly couples the activation transient (Chapter 7) to the subsequent discharge behaviour (Chapter 5) within a single simulation; the MATLAB model's empirical functional forms limit its predictive power outside the parameter range they were fitted to; no reserve battery, only standard coin cells, was fabricated.

Chapter 11

Future Work

Building on the limitations identified in Chapter 10, a few directions for future work follow naturally:

- **A correctly-formulated finite-element electrochemical model.** As discussed in Section 4.4, a more principled treatment would define the Li-SOCl_2 reaction stoichiometry from first principles using COMSOL’s general building-block physics, rather than repurposing a pre-packaged battery-chemistry interface, with particular attention to keeping the model numerically well-posed as reactant concentration approaches zero near end of discharge.
- **Coupling the activation and discharge transients.** The two-phase flow model (Chapter 7) and the discharge model (Chapters 4–5) were developed and validated independently. A natural extension is to couple them, so the electrochemical discharge model receives a time-varying, spatially evolving electrolyte distribution from the activation simulation rather than assuming instantaneous, uniform electrolyte availability.
- **LiCl precipitation and its effect on transport.** The present models treat the passivating LiCl layer’s resistive effect empirically. A more detailed treatment would model LiCl precipitation and its effect on local porosity and ionic transport explicitly.
- **Experimental validation via reserve Li-SOCl₂ cell CAD.:** So far, we have only seen the fabrication of a basic coin-cell which is a point of introduction into the cell-making procedures at ESME lab. There is indeed scope to model the artillery fuze battery in real life as well to finally produce a real life reserve battery worth application in practical scenarios.

Chapter 12

Conclusion

This project set out to investigate whether reserve battery behaviour could be computationally modelled, using Li-SOCl₂ as the target chemistry, alongside gaining practical exposure to lithium battery fabrication techniques. The results are mixed, in the way that is typical of exploratory undergraduate research, and are reported here without dressing them up.

On the achievement side: a working two-phase electrolyte activation simulation (Chapter 7); two distinct attempts to adapt COMSOL's Battery Design Module to Li-SOCl₂ chemistry, which, even though neither produced a fully satisfactory discharge curve, left a specific, evidence-grounded diagnosis of why each fell short (Chapter 4), a genuine technical contribution in its own right and not just a negative result; a complete, internally consistent lumped-parameter discharge model built independently in MATLAB and validated by direct execution (Chapter 5); a conceptual mechanical design explored in CAD (Chapter 8); and hands-on laboratory exposure to coin-cell fabrication (Chapter 9).

On the limitations side: no reserve battery was fabricated, only standard laboratory coin cells; the COMSOL electrochemical modelling did not reach a numerically stable full discharge curve for either adaptation strategy; and the MATLAB model, while stable and complete, relies on empirically-fitted rather than first-principles-derived functional forms for several key behaviours.

If there is one finding to take from this project, it is not a specific number but a diagnosis: general-purpose commercial multiphysics tools, however capable, carry chemistry-specific assumptions that are not always visible until a model is pushed outside the regime it was designed for. Recognising that, rather than trusting a converged solver output at face value, is probably the most transferable outcome of this SURGE project. The directions in Chapter 11 lay out a concrete path toward a more complete treatment, if this line of work continues.

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